

Multiobjective Optimization

Interactive Methods: Interaction with the Decision Maker

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Review of Previous Lectures

A-priori Methods - Recap

Key Concepts:

- Preferences articulated **before** solution
- Methods covered:
 - 1 Multiattribute Utility Functions
 - 2 Prior Assessment of Weights
 - 3 Goal Programming
 - 4 Compromise Programming
 - 5 Surrogate Worth Tradeoff (SWT)

Today:

- **Interactive Methods** - Preferences articulated **during** solution
- Decision maker participates in the solution process

Interactive Methods - Overview

What Are Interactive Methods?

Definition:

Methods that involve the decision maker in an iterative process, where preferences are articulated progressively as the solution evolves.

Key Characteristics:

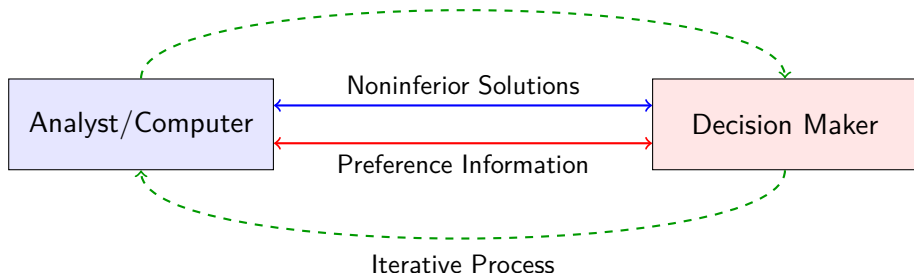
- **Bottom-up** (analyst $\rightarrow$ DM) and **top-down** (DM $\rightarrow$ analyst) flows
- Decision maker learns about tradeoffs during process
- Preferences are refined over iterations
- Generally requires computer interaction

Motivation:

"Decision makers may not know their preferences until they see alternatives."

Information Flow in Interactive Methods

Bidirectional Communication



Goal: Converge to best-compromise solution

Classification of Interactive Methods

Types of Interaction

INTERACTIVE METHODS

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graph TD; A[INTERACTIVE METHODS] --> B[Step Method (Benayoun et al., 1971)]; A --> C[SMOP (Monarchi et al., 1973)]; B --> D[Geoffrion-Dyer-Feinberg (1972)]; B --> E[Interactive Goal Programming (Dyer, 1972)]; C --> F[Belenson-Kapur (1973)]; E --> G[Zionts-Wallenius (1976)];
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Step Method (Benayoun et al., 1971)

SMOP (Monarchi et al., 1973)

Geoffrion-Dyer-Feinberg (1972)

Belenson-Kapur (1973)

Interactive Goal Programming (Dyer, 1972)

Zionts-Wallenius (1976)

Step Method - Overview

Benayoun et al. (1971)

Key Idea:

"Minimize distance from ideal, then relax objectives based on decision maker's preferences."

Core Components:

- 1 Payoff table (individual optima)
- 2 Weighted distance from ideal (d_∞ metric)
- 3 Decision maker identifies objective to relax
- 4 Iterative refinement

Claim: Converges in at most p iterations (p = number of objectives)

Step Method - Payoff Table

Initial Step

Construct Payoff Table:

Solution	Z_1	Z_2	$\dots$	Z_p
$\mathbf{x}^1$ (max Z_1)	M_1	$Z_2(\mathbf{x}^1)$	$\dots$	$Z_p(\mathbf{x}^1)$
$\mathbf{x}^2$ (max Z_2)	$Z_1(\mathbf{x}^2)$	M_2	$\dots$	$Z_p(\mathbf{x}^2)$
$\vdots$	$\vdots$	$\vdots$	$\ddots$	$\vdots$
$\mathbf{x}^p$ (max Z_p)	$Z_1(\mathbf{x}^p)$	$Z_2(\mathbf{x}^p)$	$\dots$	M_p

Where:

- M_k : Maximum value of objective k
- n_k : Minimum value of objective k in table

Step Method - Weight Calculation

Computing the Weights

Weights for Distance Metric:

$$\pi_k = \frac{\alpha_k}{\sum_{j=1}^p \alpha_j}$$

Where:

$$\alpha_k = \frac{M_k - n_k}{\left(\sum_{j=1}^n (c_j^k)^2\right)^{1/2}}$$

Interpretation:

- α_k captures variation in objective k
- Objectives with larger variation get larger weights
- Accounts for differences in scale

Step Method - Distance Problem

Minimizing Distance from Ideal

Formulation:

minimize λ

Subject to:

$$\pi_k(M_k - Z_k(\mathbf{x})) \leq \lambda \quad \forall k$$

$$\mathbf{x} \in \mathbf{F}_d$$

$$\lambda \geq 0$$

Interpretation:

- Minimizes the **maximum weighted deviation** (d_∞ metric)
- Ensures balanced solution across objectives

Step Method - Decision Maker Interaction

Relaxing Objectives

At each iteration:

- 1 Present current solution to decision maker
- 2 Ask: "Is this satisfactory?"
- 3 If **No**:
 - Identify objective k^* that is at a satisfactory level
 - Specify ΔZ_{k^*} (amount to sacrifice)
 - Add constraint: $Z_{k^*}(\mathbf{x}) \geq Z_{k^*}(\mathbf{x}^i) - \Delta Z_{k^*}$
 - Remove objective k^* from distance calculation
 - Repeat with reduced feasible region

Note: Process continues until satisfaction or p iterations

Step Method - Example

Sample Application (Cohon, 1978)

Sample problem:

$$Z_1 = 5x_1 - 2x_2, \quad Z_2 = -x_1 + 4x_2$$

Payoff Table:

Solution	Z_1	Z_2
$\mathbf{x}^1$ (max Z_1)	30	-6
$\mathbf{x}^2$ (max Z_2)	-3	15

Weights: $\pi_1 = 0.37$, $\pi_2 = 0.63$

Step Method - Example Results

Iteration Details

Iteration 0:

$$\begin{aligned} & \text{minimize } \lambda \\ & \text{s.t. } 0.37(30 - Z_1) \leq \lambda \\ & \quad 0.63(15 - Z_2) \leq \lambda \end{aligned}$$

Solution: $x_1 = 4.83$, $x_2 = 3.17$

$$Z_1 = 17.9, \quad Z_2 = 7.9$$

Decision Maker: "Insufficient Z_2 "

Relax: $\Delta Z_1 = 3$ ($Z_1 \geq 14.9$)

Step Method - Example Final

Iteration 1

New Problem:

$$\text{maximize } Z_2(x_1, x_2)$$

$$\text{s.t. } (x_1, x_2) \in \mathbf{F}_d$$

$$Z_1(x_1, x_2) \geq 14.9$$

Solution:

$$x_1 = 4.41, \quad x_2 = 3.59$$

$$Z_1 = 14.9, \quad Z_2 = 10.0$$

Decision Maker: Satisfied!

Note: If not satisfied, no solution exists (given willingness to sacrifice)

Geoffrion-Dyer-Feinberg Method

Interactive Utility Function Approximation

Key Idea:

"Use local approximations of the utility function to find direction of improvement."

Two Subproblems:

- 1 **Direction Problem:** Find best direction to improve
- 2 **Step Size Problem:** How far to move in that direction

Information Required:

- Marginal rates of substitution (MRS)
- Choice among generated alternatives

Geoffrion Method - Direction Problem

Finding the Best Direction

Direction Subproblem:

$$\begin{aligned} & \text{maximize} \quad \sum_{k=1}^p \text{MRS}_{kr}(\mathbf{x}^i) \nabla Z_k(\mathbf{x}^i) \mathbf{y} \\ & \text{s.t.} \quad \mathbf{y} \in \mathbf{F}_d \end{aligned}$$

Where:

- MRS_{kr} : Marginal rate of substitution between Z_k and Z_r
- Z_r : Reference objective (numeraire)
- $\mathbf{x}^i$: Current solution

Result: Direction $\mathbf{d}^i = \mathbf{y}^i - \mathbf{x}^i$

Interpretation: Direction of steepest ascent in utility space

Geoffrion Method - Step Size Problem

How Far to Move

Step Size Subproblem:

$$\begin{aligned} &\text{maximize } U[Z_1(\mathbf{x}^i + t\mathbf{d}^i), \dots, Z_p(\mathbf{x}^i + t\mathbf{d}^i)] \\ &\text{s.t. } 0 \leq t \leq 1 \end{aligned}$$

Practical Implementation:

- 1 Generate several solutions along direction
- 2 Present to decision maker
- 3 Decision maker chooses preferred solution

Result: t^* from decision maker's choice

$$\mathbf{x}^{i+1} = \mathbf{x}^i + t^*\mathbf{d}^i$$

Assumed Utility Function:

$$U(Z_1, Z_2) = Z_1 \cdot Z_2$$

Starting Point: $\mathbf{x}^0 = (6, 2)$

$$Z_1 = 26, \quad Z_2 = 2, \quad U = 52$$

MRS at $\mathbf{x}^0$:

$$\text{MRS}_{21} = -\frac{\partial Z_1 / \partial Z_2}{\partial Z_2 / \partial Z_1} = \frac{26}{2} = 13$$

Geoffrion Method - Example Iterations

Convergence Process

Iteration	x_1	x_2	Z_1	Z_2
0	6.0	2.0	26.0	2.0
1	4.2	2.7	15.6	6.6
2	4.0	4.0	12.0	12.0
3	4.3	3.7	14.4	10.3
4	4.3	3.7	14.4	10.3

Convergence: $\mathbf{x}^4 = \mathbf{x}^3$ (stopping criterion met)

Best-Compromise: $(Z_1^*, Z_2^*) = (14.4, 10.3)$

Interactive Goal Programming

Dyer's Method (1972)

Key Idea:

"Combine goal programming with interactive preference elicitation."

Procedure:

- 1 Decision maker specifies goals and weights
- 2 Solve goal programming problem
- 3 Present solution and tradeoffs
- 4 Decision maker can modify goals/weights
- 5 Repeat until satisfied

Advantage:

- Decision maker can learn from tradeoffs
- Preferences can be adjusted iteratively

Zionts-Wallenius Method

Interactive Simplex-Based Method

Key Idea:

"Use simplex pivots to generate adjacent noninferior solutions, let decision maker choose."

Procedure:

- 1 Start with arbitrary noninferior solution
- 2 Identify adjacent noninferior solutions
- 3 Ask decision maker if they prefer any
- 4 Move to preferred solution
- 5 Repeat until no preference found

Advantages:

- Only local preference information needed
- Handles both linear and nonlinear preferences

Empirical Results (Wallenius, 1975)

Testing Interactive Methods

Experiment:

- Subjects: Business students and managers
- Methods tested:
 - 1 Step Method
 - 2 Geoffrion-Dyer-Feinberg Method
 - 3 Trial-and-error (unstructured)
- Subjects ranked methods on ease, confidence, preference

Key Finding:

"The two formal interactive procedures were found to be no better, and in many cases worse, than the unstructured trial-and-error methods."

Implication: Generating methods may be more natural for decision makers

Comparison of Interactive Methods

Summary

Method	Information Needed	Convergence	Complexity
Step	Objective to relax	$\leq p$ iterations	Low
Geoffrion	MRS, step choice	Variable	High
Interactive GP	Goals, weights	Variable	Medium
Zionts-Wallenius	Preference among adjacent solutions	Variable	Medium

Key Insight:

"Interactive methods attempt to help decision makers discover their preferences, but may not always be more effective than simple exploration."

Summary

Key Takeaways

Interactive Methods:

- Preferences articulated **during** the solution process
- **Bidirectional** information flow (DM $\leftrightarrow$ Analyst)
- Decision maker learns about tradeoffs
- Solutions converge to best-compromise

Key Methods:

- 1 **Step Method:** Distance from ideal, then relax objectives
- 2 **Geoffrion-Dyer-Feinberg:** Local utility approximation
- 3 **Interactive GP:** Adjustable goals and weights
- 4 **Zionts-Wallenius:** Simplex-based local search

Next: Multiple-Decision-Maker Methods

Topics:

- ① Aggregation of Individual Preferences
 - Welfare economics
 - Social choice theory
 - Arrow's impossibility theorem
- ② Counseling Individual Decision Makers
- ③ Predicting Political Outcomes
 - Paretian analysis
 - Game theory
 - Vote-trading models

Key Question: How do we handle multiple conflicting decision makers?



COHON, J. L. *Multiobjective Programming and Planning*. Dover Publications, 2013.